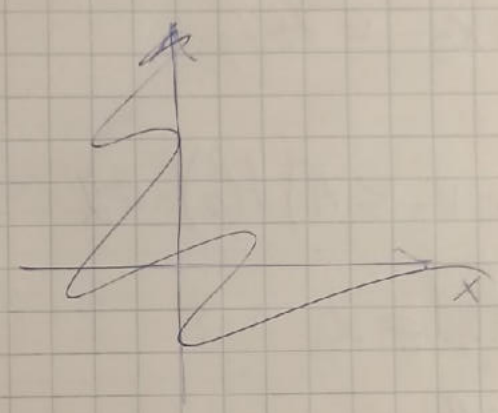
T 11.

$H_0: \; p_0(x) = 1\{(0,1)\}$

$H_1: \; p_1(x) = \frac{e^{1-x}}{e-1}\{(0,1)\}$

$$l(x) = \frac{L_1(x)}{L_0(x)} = \frac{e^{1-x_1}}{(e-1)\cdot 1} \geq C$$

$$x_1 \leq A$$



$$P(x_1 \leq A \mid H_0) = \alpha$$

$$\int_0^A dx = \alpha$$

$$A = \alpha;$$

$G: \; x_1 \leq \alpha$

$\alpha_1 = \alpha$

$$W = P(x_1 \leq A \mid H_1) = \int_0^A \frac{e^{1-x}}{e-1} dx =$$

$$= \frac{e}{e-1} \cdot (-e^{-x}) \Big|_0^A = \frac{e}{e-1}\left(1 - e^{-A}\right) =$$

$$= \frac{e}{e-1}\left(1 - e^{-\alpha}\right)$$

$$\cancel{W = 1 - \alpha_2 = 1 - \frac{e}{e-1} + \frac{e}{e-1} e^{-\alpha} = \frac{1}{e-1}\left(e^{-\alpha+1} - 1\right)}$$

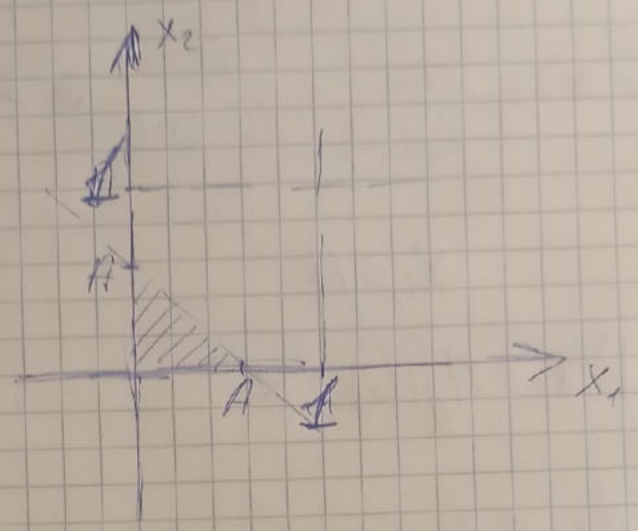
$$\alpha_2 = 1 - W$$

б) $\underline{n=2}$

$$\ell(\vec{x}_2) = \frac{L_1(\vec{x}_2)}{L_0(\vec{x}_2)} = \frac{e^{1-x_1}e^{1-x_2}}{(e-1)^2} =$$

$$= \frac{e^{2-x_1-x_2}}{(e-1)^2} \geq C$$

$$x_1+x_2 \leq A$$



$$P(x_1+x_2 \leq A \mid H_0) = \alpha$$

$$\iint\limits_{x_1+x_2\leq A} dx_1 dx_2 = \alpha$$

$$\int_0^A dx_1 \int_0^{A-x_1} dx_2 = \int_0^A (A-x_1)\,dx_1 = A^2 - \frac{A^2}{2} = \frac{A^2}{2} = \alpha \Rightarrow$$

$$\Rightarrow A = \sqrt{2\alpha}$$

При этом $A \leq 1 \Rightarrow$

$\Rightarrow \alpha \leq \frac{1}{2}$

(что нормально)

$G: x_1+x_2 \leq \sqrt{2\alpha}$

(независимы)

$\alpha_1 = \alpha$

$$W = \iint\limits_{x_1+x_2\leq A} \frac{e^{2-x_1-x_2}}{(e-1)^2}\,dx_1 dx_2 =$$

$$= \frac{e^2}{(e-1)^2} \int_0^A dx_1 \int_0^{A-x_1} e^{-x_1-x_2}\,dx_2 =$$

$$= \frac{e^2}{(e-1)^2} \int_0^A e^{-x_1}dx_1 \int_0^{A-x_1} e^{-x_2}dx_2 \longrightarrow$$

$$\Leftarrow \left(\frac{e}{e-1}\right)^2 \int_0^A e^{-x_1} dx_1 \cdot \left(1 - e^{+x_1 - A}\right) =$$

$$= \left(\frac{e}{e-1}\right)^2 \int_0^A \left(e^{-x_1} - e^{-A}\right) dx_1 = \left(\frac{e}{e-1}\right)^2 \cdot \left(1 - e^{-A} - Ae^{-A}\right)$$

$$\alpha_2 = 1 - W$$

c) $\underline{\text{III}}$

$$\ell(\vec{X_n}) = \frac{L_1(\vec{X_n})}{L_0(\vec{X_n})} = \frac{e^{n - 2x_i}}{(e-1)^n} \geq C$$

$$\sum_{i=1}^{n} x_i \leq B$$

$$\bar{x} \leq A$$

x_i однак. распред., независимы;

$$Mx_i = \int_0^1 x\,dx = \frac{1}{2}$$

$$Dx_i = \int_0^1 x^2 dx - \frac{1}{4} = \frac{1}{3} - \frac{1}{4} = \frac{1}{12}$$ (огранич. мат. мож.) ← при условии H_0

$\Rightarrow$ справедлива простейшая Ц.П.Т. и

$$\frac{\bar{X} - M\xi}{\sqrt{D\xi}} \sqrt{n} \sim N(0,1)$$

$\downarrow$

$$\bar{X} \le A \Rightarrow \frac{\bar{X} - M\xi}{\sqrt{D\xi}}\sqrt{n} \le \frac{A - M\xi}{\sqrt{D\xi}}\sqrt{n}$$

$$\left(\bar{X} - \frac{1}{2}\right)\sqrt{12n} \le \left(A - \frac{1}{2}\right)\sqrt{12n}$$

$$P\left(\left(\bar{X} - \frac{1}{2}\right)\sqrt{12n} \le \underbrace{\left(A - \frac{1}{2}\right)\sqrt{12n}}_{= u_\alpha} \,\middle|\, H_0\right) = \alpha$$

$$\cancel{\left(\bar{X} - \frac{1}{2}\right)\sqrt{12n} \le u_\alpha}$$

$$\left(A - \frac{1}{2}\right)\sqrt{12n} = u_\alpha$$

$$A = \frac{u_\alpha}{\sqrt{12n}} + \frac{1}{2}$$

$$G: \bar{X} \le \frac{u_\alpha}{\sqrt{12n}} + \frac{1}{2}$$

$$\alpha_1 = \alpha_2$$

$$W = P\left(\bar{X} \le \frac{u_\alpha}{\sqrt{12n}} + \frac{1}{2} \,\middle|\, H_1\right)$$

$$M\xi = \int_0^1 x \cdot \frac{e^{1-x}}{e-1}\,dx = \frac{e}{e-1}\int_0^1 x e^{-x}\,dx = \left(\frac{e}{e-1}\right)\left(-xe^{-x}\Big|_0^1 + \int_0^1 e^{-x}\,dx\right) =$$

$$= \frac{e}{e-1}\left(-e^{-1} + \cancel{1} - e^{-1}\right) = \frac{e(1-2e^{-1})}{e-1} = \frac{e-2}{e-1}$$

$$M\xi^2 = \frac{e}{e-1}\int_0^1 x^2 e^{-x}\,dx$$

$$\int_0^1 x^2 e^{-x}\,dx = -x^2 e^{-x}\Big|_0^1 + \int_0^1 2xe^{-x}\,dx = -e^{-1} - 2xe^{-x}\Big|_0^1 +$$

$$+ \int_0^1 2e^{-x}\,dx = \cancel{e} \rightarrow$$

$$\Theta = -e^{-1} - 2e^{-1} - 2(-1+e^{-1}) = 2 - \frac{5}{e}$$

$$\cancel{D\xi = 2 - \frac{5}{e}}$$

$$M\xi^2 = \left(\frac{e}{e-1}\right)\left(2 - \frac{5}{e}\right) = \frac{2e-5}{e-1}$$

при условии H_1

$$M\xi = \frac{e-2}{e-1} = 0{,}418$$

$$D\xi = \frac{2e-5}{e-1} - \left(\frac{e-2}{e-1}\right)^2 = \frac{2e^2-5e-2e+5-e^2+4e-4}{(e-1)^2} =$$

$$= \frac{e^2-3e+1}{(e-1)^2}$$

$$W = P\left(\frac{\bar{x}-M\xi}{\sqrt{D\xi}}\sqrt{n} \le \overbrace{\frac{A-M\xi}{\sqrt{D\xi}}\sqrt{n}}^{=\gamma} \,\middle|\, H_1\right) = \int_{-\infty}^{\gamma} \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}\,dx$$

$$\alpha_2 = 1 - W$$

Состоятельность:

$W \xrightarrow[n\to+\infty]{} 1$ $\forall$ распред. из H_1:

$$\gamma = \frac{A - M\xi}{\sqrt{D\xi}}\sqrt{n}\ ; \quad A = \frac{u_\alpha}{\sqrt{12n}} + \frac{1}{2}$$

$$\gamma = \frac{\frac{1}{2} + \frac{u_\alpha}{\sqrt{12n}} - 0{,}418}{\sqrt{D\xi}}\sqrt{n} = \frac{\frac{u_\alpha}{12} + 0{,}082\sqrt{n}}{\sqrt{D\xi}} \xrightarrow[n\to\infty]{} +\infty \Rightarrow$$

$\Rightarrow W \xrightarrow[n\to\infty]{} 1 \Rightarrow$ состоятельный

d) G: $x_{min} < c$; $c \in [0,1]$, т.к. при $c > 1$ $\alpha_1 = 1$
$c < 0$ $\alpha_1 = 0$ не интересно

$\alpha_1 \le \alpha$

$$\alpha_1 = P(x_{min} < c \mid H_0) = 1 - (1 - F(x))^n \; ; \; F(x) = \begin{cases} 0, & x \le 0 \\ x, & x \in (0,1) \\ 1, & x \ge 1 \end{cases}$$

$$\begin{cases} \alpha_1 = 1 - (1-c)^n \le \alpha \quad (c \in [0,1]) \\ \alpha_2 \longrightarrow \min \text{ (наша задача)} \end{cases}$$

$$\alpha_2 = 1 - W = 1 - P(x_{min} < c \mid H_1)$$

$$\int_0^c \frac{e^{1-x}}{e-1} dx = \frac{e}{e-1}(1 - e^{-x})$$

$$\Phi(x) = \begin{cases} 0, & x \le 0 \\ \frac{e}{e-1}(1 - e^{-x}), & x \in (0,1) \\ 1, & x > 1 \end{cases}$$

$$W = 1 - \frac{e}{e-1}(1 - \Phi(x))^n = 1 - \left(1 - \frac{e}{e-1} + \frac{e^{1-x}}{e-1}\right)^n$$

$$\alpha_2 = \left(\frac{e^{1-c} - 1}{e-1}\right)^n \xrightarrow[c=?]{} \min$$

d.1) Безусловная оптимизация:

$$\alpha_1 < \alpha \; : \; (1 - (1-c)^n) < \alpha$$

$$\left(\frac{e^{1-c} - 1}{e-1}\right)^n \to \min$$

$\downarrow$

$$\left(\left(\frac{e^{1-c}-1}{e-1}\right)^n\right)' = 0$$

$$n \cdot \left(\frac{e^{1-c}-1}{e-1}\right)^{n-1} \cdot \frac{e^{1-c}}{e-1}(-1) = 0 \Rightarrow c = 1$$ – стац. точка.

При $c = 1$ $\alpha \geq 1$ – не интересно,

(но это и в правду точка минимума)

д.г) условная оптимизация:

$$\begin{cases} 1-(1-c)^n = \alpha \\ \alpha_2 \to \min \end{cases}$$

$$c = -\sqrt[n]{1-\alpha} + 1$$

$$L = \left(\frac{e^{1-c}-1}{e-1}\right)^n + \lambda \cdot \left(1-(1-c)^n - \alpha\right);$$

$$\begin{cases} \frac{\partial L}{\partial c} = 0 \\ 1-(1-c)^n - \alpha = 0 \end{cases} \to c = 1-\sqrt[n]{1-\alpha} \; ; \; (1-c) = \sqrt[n]{1-\alpha}$$

$$\frac{\cancel{n}}{(e-1)^n} \cdot (e^{1-c}-1)^{n-1}(-e^{1-c}) + \lambda \cancel{n} (1-c)^n = 0$$

$$\lambda = \frac{(e^{1-c}-1)^{n-1} e^{1-c}}{(e-1)^n (1-c)^n} = \frac{(e^{-1-c}-1)^{n-1} e^{1-c}}{(e-1)^n (1-\alpha)}$$

$$d^2L = \frac{\partial^2 L}{\partial c^2} dc^2 = -\frac{n}{(e-1)^n} \cdot \left((e^{1-c}-1)^{n-1}(e^{1-c})\right)' dc^2 =$$

$$= \frac{-n}{(e-1)^n}\Big((n-1)(e^{1-c}-1)^{n-2}\cdot(-e^{1-c})\,e^{1-c} +$$

$$+ (-e^{1-c})(e^{1-c}-1)^{n-1}\Big)\,dc^2 =$$

$$= \frac{+n}{(e-1)^n}(e^{1-c}-1)^{n-2}\left((n-1)(e^{2-2c}) + e^{1-c}(e^{1-c}-1)^{n-1}\right)dc^2 \to 0 \Rightarrow$$

точка условного минимума

{здесь везде $1-c = \sqrt[n]{1-\alpha}$}

$$dg = d\left(1-(1-c)^n-\alpha\right) = -d(1-c)^n = n(1-c)^{n-1}dc$$

Итого:

$G: x_{min} < 1-\sqrt[n]{1-\alpha}$

$\alpha_1 = \alpha$ (из усл. оптимиз.)

$$\alpha_2 = \left(\frac{e^{\sqrt[n]{1-\alpha}}-1}{e-1}\right)^n$$

$W = 1-\alpha_2$

d, 3) Состоятельность:

$$\sqrt[n]{1-\alpha} = e^{\frac{1}{n}\ln(1-\alpha)} = 1 + \frac{1}{n}\ln(1-\alpha) + o\left(\frac{1}{n}\right)$$

$$e^{\sqrt[n]{1-\alpha}} = e^{1+\frac{1}{n}\ln(1-\alpha)+o\left(\frac{1}{n}\right)} = e^1\left(1+\frac{1}{n}\ln(1-\alpha)+o\left(\frac{1}{n}\right)\right)$$

$$\lim_{n\to\infty}\left(\frac{e\left(1+\frac{1}{n}\ln(1-\alpha)+o\left(\frac{1}{n}\right)\right)-1}{e-1}\right)^n=$$

$$=\lim_{n\to\infty}\left(1+\frac{e}{n}\cdot\frac{\ln(1-\alpha)}{e-1}+\frac{e}{e-1}\cdot o\left(\frac{1}{n^2}\right)\right)^n=$$

$$=e^{\frac{\ln(1-\alpha)e}{e-1}}=(1-\alpha)^{\frac{e}{e-1}}\xrightarrow[n\to\infty]{}0 \Rightarrow \text{не явл. сост.}$$

Т12.

H_0: $p_4 = \frac{1}{3}$; $p_3 = \frac{1}{6}$; $p_2 = p_1 = \frac{1}{4}$

H_1: $p_1 = p_2 = p_3 = p_4 = \frac{1}{4}$

$$l(\vec{x_i}) = \frac{L_1(\vec{x_i})}{L_0(\vec{x_i})} = \frac{1}{16 p_a p_b} \geq C \quad ; \quad \underline{\alpha = 0{,}2}$$

Число C уменьшаем до тех пор, пока

$$P(l \geq C | H_0) \leq \alpha$$

$$l_{max} = l_{(1)} = \frac{36}{16} = \frac{9}{4} \quad (a = b = 3)$$

$$P_{0(1)} = \frac{1}{36} \qquad P(l \geq C | H_0) = \frac{1}{36} \leq 0{,}2$$

$$P_{1(1)} = \frac{1}{16}$$

номер гипотезы — номер k-го максимума l

$$l_{(2)} = \frac{24}{16} = \frac{3}{2} \qquad \begin{pmatrix} a = 3;\ b \in (1,2) \\ a \in (1,2);\ b = 3 \end{pmatrix}$$

$$P_{0(2)} = \frac{1}{6} \cdot \left(\frac{1}{4} + \frac{1}{4}\right) + \left(\frac{1}{4} + \frac{1}{4}\right) \cdot \frac{1}{6} = \frac{1}{6}$$

$$P_{1(2)} = \frac{1}{4}\left(\frac{1}{4} + \frac{1}{4}\right) + \left(\frac{1}{4} + \frac{1}{4}\right) \cdot \frac{1}{4} = \frac{1}{4}$$

$$P(l \geq C | H_0) = \frac{1}{36} + \frac{1}{6} = \frac{7}{36} \leq 0{,}2$$

$$l_{(3)} = \frac{16}{16} = 1 \qquad \left(a \in (1,2);\ b \in (1,2)\right)$$

$$P_{0(3)} = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$$

$$P_{1(3)} = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$$

$$P(l \geq C | H_0) = \frac{1}{36} + \frac{1}{6} + \frac{1}{4} > 0{,}2 \Rightarrow$$

$\Rightarrow$ подходит $\underline{C = \frac{3}{2}}$

$G: \ \ell \geq \frac{3}{2}$

$\alpha_1 = P(\ell \geq C | H_0) = \frac{7}{36} = 0{,}1944$

$W = P(\ell \geq C | H_1) = P_{1(1)} + P_{1(2)} = \frac{1}{16} + \frac{1}{4} = \frac{5}{16} = 0{,}3125$

$\alpha_2 = 1 - W = 0{,}6875.$

T13.

$H_0: \ \sigma_1^2 = \sigma_2^2$

$H_1: \ \sigma_1^2 \neq \sigma_2^2$

с обеих сторон.

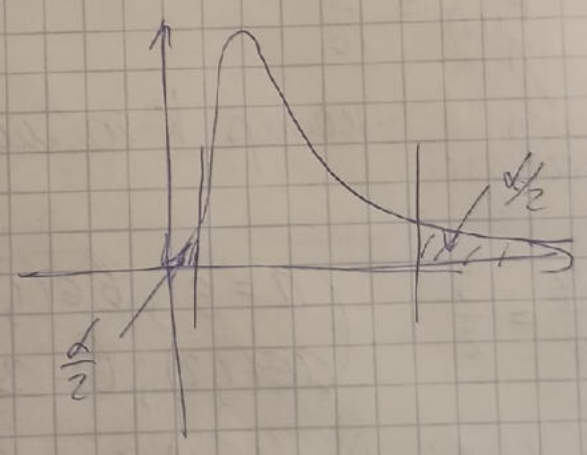
а) длина:

$h_1 \sim N(a, \sigma_1^2)$

$h_2 \sim N(b, \sigma_2^2)$

$\frac{(n-1)S_1^2}{\sigma_1^2} \sim \chi^2(n-1);$

$\frac{(m-1)S_2^2}{\sigma_2^2} \sim \chi^2(m-1);$



$\alpha = 0{,}05$

$\Delta = \frac{S_1^2/\sigma_1^2}{S_2^2/\sigma_2^2} \sim F(n-1, m-1)$

$P(\Delta \leq C | H_0) = \frac{\alpha}{2}$

$C = F_{\frac{\alpha}{2}}(n-1, m-1);$

$G: \tilde{Z} \in [0, F_{\frac{\alpha}{2}}(n-1, m-1)) \cup (F_{1-\frac{\alpha}{2}}(n-1, m-1), +\infty)$

$Z = \frac{S_1^2}{S_2^2}$ $(\sigma_1^2 = \sigma_2^2)$

$S_1^2 = \frac{n}{n-1} \cdot \tilde{\sigma}_1^2$; $S_2^2 = \frac{m}{m-1} \tilde{\sigma}_2^2$

$Z = \frac{n(m-1)\tilde{\sigma}_1^2}{(n-1)m\tilde{\sigma}_2^2} = \frac{139 \cdot 999 \cdot 5{,}722^2}{138 \cdot 1000 \cdot 6{,}161^2} = 0{,}868$

$F_{\frac{\alpha}{2}}(138, 999) = 0{,}767$

$F_{1-\frac{\alpha}{2}}(138, 999) = 1{,}272$

$\tilde{Z} \notin G \Rightarrow$ нельзя отвергнуть нулевую гипотезу.
(Мощность в конце задачи)

б) ширина:

$H_0: \sigma_1^2 = \sigma_2^2$ $\quad n = 139 \quad \tilde{\sigma}_1^2 = 4{,}612$

$H_1: \sigma_1^2 \neq \sigma_2^2$ $\quad m = 1000 \quad \tilde{\sigma}_2^2 = 5{,}055$

$\omega_1 \sim N(a, \sigma_1^2)$ $\quad \alpha = 0{,}05$

$\omega_2 \sim N(b, \sigma_2^2)$

По теор. Фишера:

$$\frac{S_1^2(n-1)}{\sigma_1^2} \sim \chi^2(n-1); \quad \frac{S_2^2(m-1)}{\sigma_2^2} \sim \chi^2(m-1)$$

$$\Delta = \frac{\left(\frac{S_1^2}{\sigma_1^2}\right)}{\left(\frac{S_2^2}{\sigma_2^2}\right)} \sim F(n-1, m-1)$$

$$S_1^2 = \frac{n}{n-1} \cdot \tilde{\sigma}_1^2 \qquad S_2^2 = \frac{m}{m-1} \tilde{\sigma}_2^2$$

по выборке

$$P(\tilde{\Delta} \leq \underset{\underset{F_{\frac{\alpha}{2}}(n-1, m-1)}{\parallel}}{C_1} | H_0) = \frac{\alpha}{2}; \quad P(\tilde{\Delta} \geq \underset{\underset{F_{1-\frac{\alpha}{2}}(n-1, m-1)}{\parallel}}{C_2} | H_0) = \frac{\alpha}{2}$$

$$\tilde{\Delta} = \frac{S_1^2}{S_2^2} \quad (\text{т.к. при } H_0 \ \sigma_1^2 = \sigma_2^2)$$

$$G: \tilde{\Delta} \in \left(-\infty; F_{\frac{\alpha}{2}}(n-1, m-1)\right) \cup \left(F_{1-\frac{\alpha}{2}}(n-1, m-1); +\infty\right)$$

$$\tilde{\Delta} = \frac{n\cdot(m-1)}{m(n-1)}\cdot\frac{\tilde{\sigma}_1^2}{\tilde{\sigma}_2^2} = \frac{139\cdot 999}{138\cdot 1000}\cdot\frac{4{,}612^2}{5{,}055^2} = 0{,}8376$$

$F_{\frac{\alpha}{2}}(n-1, m-1) = 0{,}767$

$F_{1-\frac{\alpha}{2}}(n-1, m-1) = 1{,}272$

$\tilde{\Delta} \notin G \Rightarrow$ нельзя отвергнуть $H_0 \Rightarrow$ можно считать расхожд. случайными

(Также можно проверить только левую или только правую границы: $F_\alpha = 0{,}8012$;
$F_{1-\alpha} = 1{,}2239$;
тоже $\tilde{\Delta} \notin G$ в случае длины и ширины)

$W = P(\tilde{\Delta} \in G \mid H_1)$

$$W = P(\tilde{\Delta} \in G \mid H_1) = P\left(\tilde{\Delta} \le F_{\frac{\alpha}{2}}(n-1, m-1) \mid H_1\right) +$$

$$+ P\left(\tilde{\Delta} \ge F_{1-\frac{\alpha}{2}}(n-1, m-1) \mid H_1\right)$$

$$W = P\left(\frac{S_1^2}{S_2^2} \le F_{\frac{\alpha}{2}}(n-1, m-1) \mid H_1\right) +$$

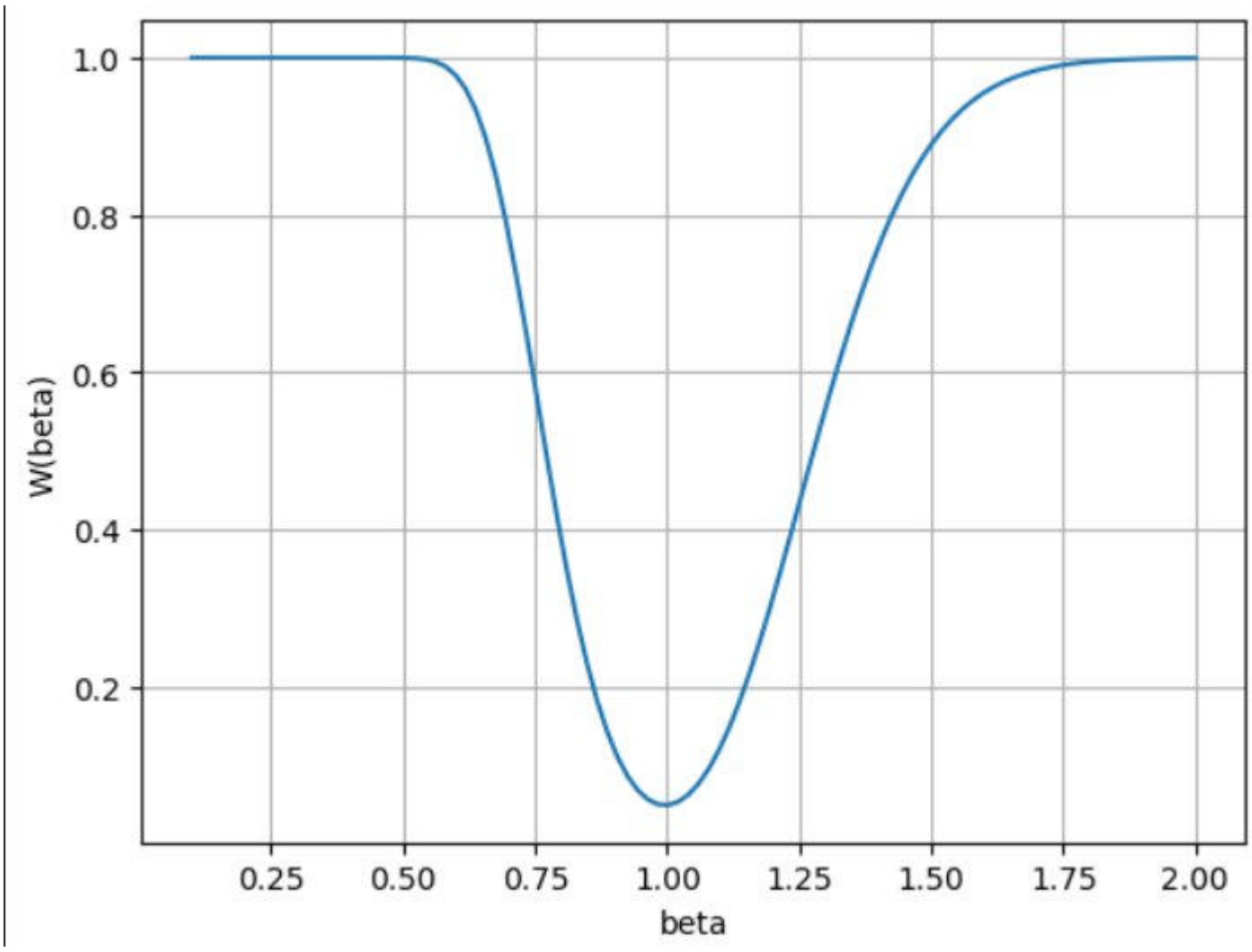
$$+ P\left(\frac{S_1^2}{S_2^2} \ge F_{1-\frac{\alpha}{2}}(n-1, m-1) \mid H_1\right)$$

имеют неизв. распред, но если нормируем на $\beta = \frac{\sigma_1^2}{\sigma_2^2}$, то будет <u>Фишер.</u>

$$W = P\left(\frac{S_1^2}{S_2^2}\frac{1}{\beta} \le \frac{F_{\frac{\alpha}{2}}(n-1, m-1)}{\beta}\right) + P\left(\frac{S_1^2}{S_2^2}\frac{1}{\beta} \ge \frac{F_{1-\frac{\alpha}{2}}(n-1, m-1)}{\beta}\right)$$

$$= \int_{-\infty}^{\gamma} F'_{(n-1, m-1)}(x)\,dx + \int_{\rho}^{+\infty} F'_{(n-1, m-1)}(x)\,dx$$

здесь можно написать ноль


1.0
0.8
0.6
0.4
0.2
W(beta)
0.25
0.50
0.75
1.00
1.25
1.50
1.75
2.00
beta

Т14.

$\sigma_x^2 = 2$ $\quad$ $\alpha = 0{,}05$

$\sigma_y^2 = 1$

$\xi \sim N(a, \sigma_x^2)$; $y \sim N(b, \sigma_y^2)$

$x_3 = \{-1{,}11; -6{,}10; 3{,}42\}$; $y_2 = \{-3{,}29, -2{,}91\}$

$H_0: a = b$ $\quad$ $\bar{x} = -1{,}597$

$H_1: a > b$ $\quad$ $\bar{y} = -2{,}6$

По т. Фишера:

$$\frac{\bar{x} - a}{\sigma_x}\sqrt{n} \sim N(0,1) \qquad \frac{\bar{y} - b}{\sigma_y}\sqrt{m} \sim N(0,1)$$

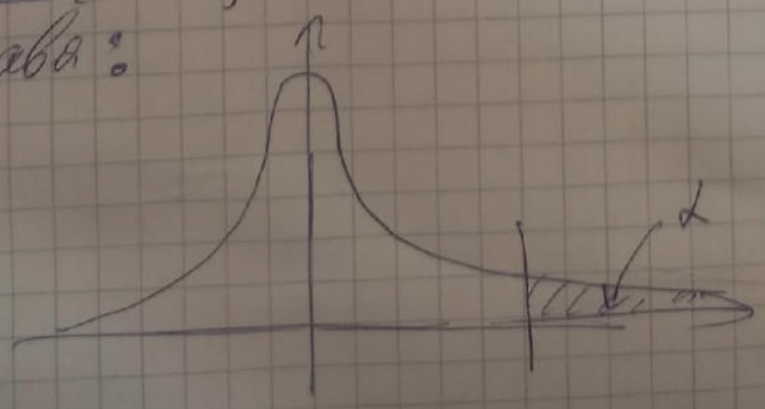
$$\bar{x} - a \sim N\left(0, \frac{\sigma_x^2}{n}\right) \qquad \bar{y} - b \sim N\left(0, \frac{\sigma_y^2}{m}\right)$$

$$\bar{x} - a - (\bar{y} - b) \sim N\left(0, \frac{\sigma_x^2}{n} + \frac{\sigma_y^2}{m}\right)$$

независимы

$$\Delta = \frac{\bar{x} - \bar{y} - a + b}{\sqrt{\frac{\sigma_x^2}{n} + \frac{\sigma_y^2}{m}}} \sim N(0,1)$$

~~если $a > b$, то~~ при H_0 критическая область справа:



$$P(\tilde{\Delta} \geq C \mid H_0) = \alpha$$

$$C = N_{1-\alpha} = 1{,}645$$

$$\tilde{\Delta} = \frac{\bar{x} - \bar{y}}{\sqrt{\frac{\sigma_x^2}{n} + \frac{\sigma_y^2}{m}}} = \frac{-1{,}587 + 2{,}6}{\sqrt{\frac{2}{3} + \frac{1}{2}}} = 0{,}9286$$

$G: \tilde{\Delta} \geq 1{,}645.$

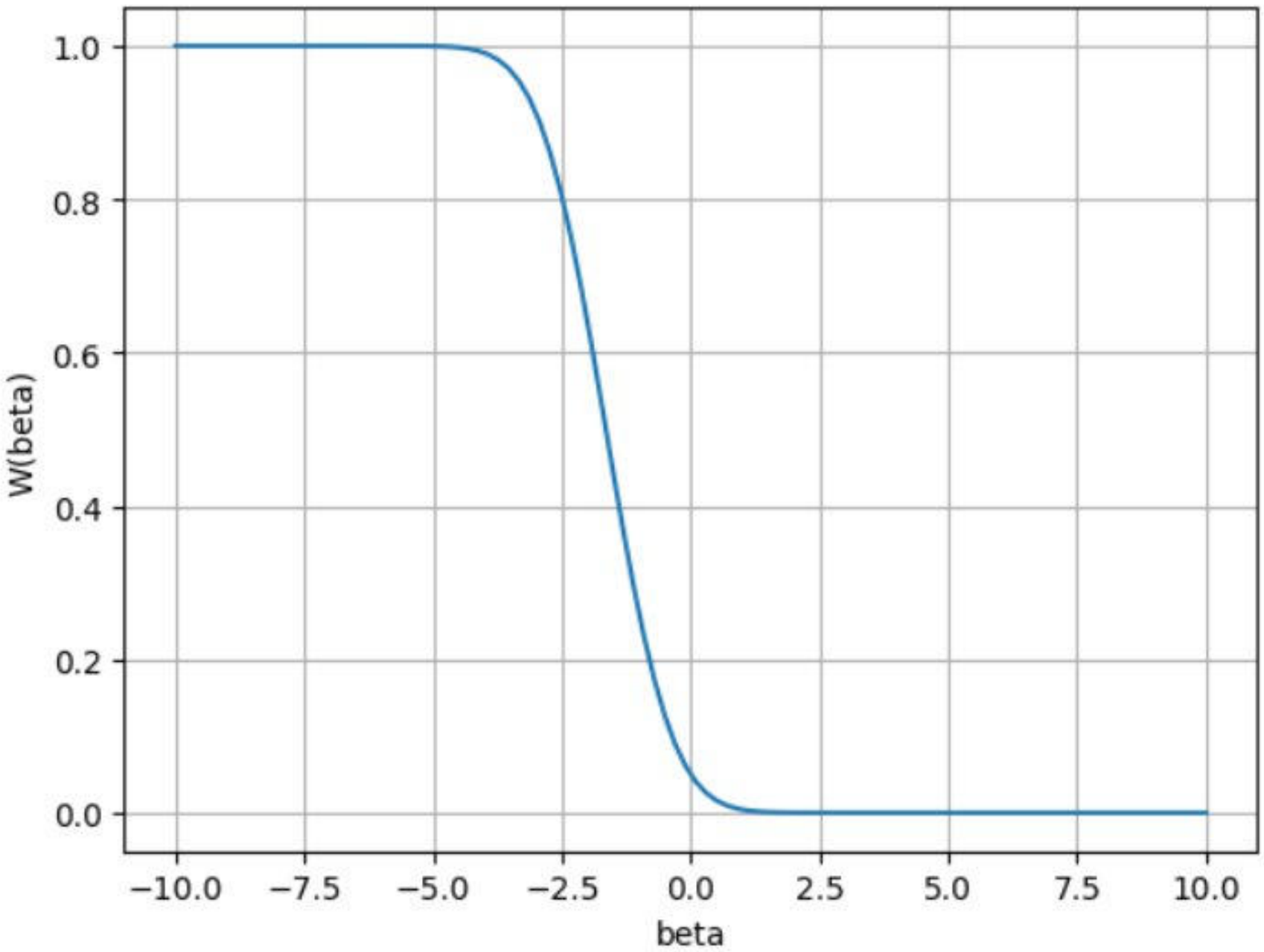
$\tilde{\Delta} \notin G \Rightarrow$ нельзя отвергнуть H_0

$$W = P(\tilde{\Delta} \geq C \mid H_1) =$$

$$= P\left(\underbrace{\frac{\bar{x} - \bar{y}}{\sqrt{\frac{\sigma_x^2}{n} + \frac{\sigma_y^2}{m}}} + \frac{b-a}{\sqrt{\frac{\sigma_x^2}{n} + \frac{\sigma_y^2}{m}}}}_{\text{по Т Фишера имеет } N(0,1) \text{ распред.}} \geq C + \frac{b-a}{\sqrt{\frac{\sigma_x^2}{n} + \frac{\sigma_y^2}{m}}}\right)$$

$$\frac{b-a}{\sqrt{\frac{\sigma_x^2}{n} + \frac{\sigma_y^2}{m}}} = \beta$$

$$\underset{W(\beta)}{\overset{\shortparallel}{W}} = P\left(\frac{\bar{x} - \bar{y}}{\sqrt{\frac{\sigma_x^2}{n} + \frac{\sigma_y^2}{m}}} + \beta \geq C + \beta\right) = \int_{\underset{1{,}645}{\overset{\shortparallel}{C+\beta}}}^{+\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} dx$$


1.0
0.8
0.6
0.4
0.2
0.0
W(beta)
−10.0
−7.5
−5.0
−2.5
0.0
2.5
5.0
7.5
10.0
beta